The documentation and process conversion measures necessary to comply with this document shall be completed by 21 April 2010.

INCH-POUND

MIL-PRF-19500/727B
21 January 2010
SUPERSEDING
MIL-PRF-19500/727A
15 April 2008

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, SWITCHING,
TYPES 2N5010 THROUGH 2N5015, 2N5010S THROUGH 2N5015S, 2N5010U4 THROUGH 2N5015U4,
JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN, silicon, transistors. Four levels of product assurance are provided for each encapsulated device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (TO-5 and TO-39), and 2 (U4).

* 1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

Types	I_C	I_B	V_{CBO}	V_{CER}	V_{EBO}	T_J and T_{STG} (1)
	mA dc	mA dc	V dc	V dc	V dc	°C
2N5010, 2N5010S, U4	200	20	500	500	5	-65 to +200
2N5011, 2N5011S, U4	200	20	600	600	5	-65 to +200
2N5012, 2N5012S, U4	200	20	700	700	5	-65 to +200
2N5013, 2N5013S, U4	200	20	800	800	5	-65 to +200
2N5014, 2N5014S, U4	200	20	900	900	5	-65 to +200
2N5015, 2N5015S, U4	200	20	1000	1000	5	-65 to +200

Types	P_T T_A = +25°C (2) (3)	P_T T_C = +25°C (2) (3)	$R_{\theta JA}$ (3) (4)	$R_{\theta JC}$ (3) (4)
	W	W	°C/W	°C/W
2N5010 through 2N5015 and 2N5010S through 2N5015S	1	7	175	20
2N5010U4 through 2N5015U4	1	7	175	7

(1) For derating, see figures 3, 4, and 5.
(2) See figure 5 for special bias considerations under ambient operation.
(3) For thermal impedance curves, see figures 6, 7, 8, and 9.
(4) T_A = 25°C for (U4) on printed circuit board (PCB), PCB = FR4 .0625 inch (1.59 mm) 1-layer 1-Oz Cu, horizontal, still air, pads (U4) = .067 x .065 inch (1.7 x 1.65 mm) for the two smaller pads and .165 x .135 inch (4.19 x 3.43 mm) for the larger collector pad; $R_{\theta JA}$ with a defined PCB thermal resistance condition included.

* Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil .

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* 1.4 Primary electrical characteristics at T_A = +25°C. Unless otherwise specified, T_A = +25°C.

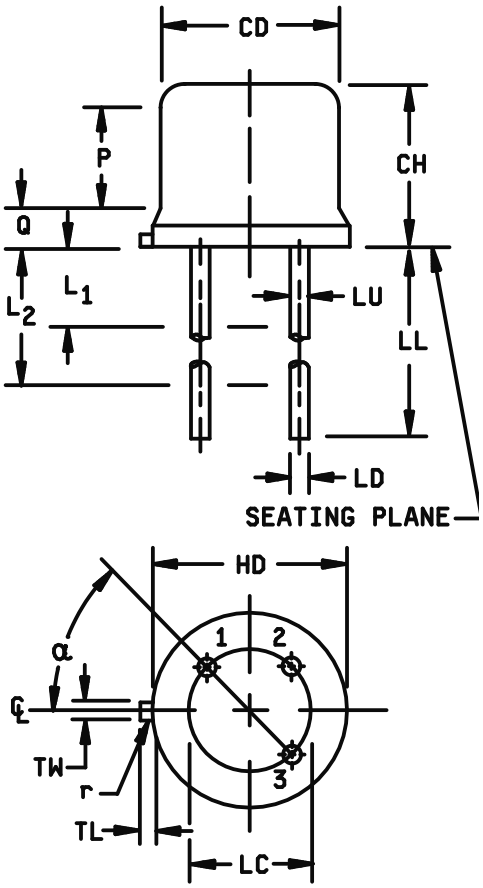
Types	h_{FE} at V_{CE} = 10 V dc				$V_{CE(sat)1}$ (1) I_C = 25 mA dc I_B = 5 mA dc	$V_{BE(sat)1}$ (1) I_C = 25 mA dc I_B = 5 mA dc	$\lvert h_{fe} \rvert$ f = 10 MHz	C_{obo} 100 kHz ≤ f ≤ 2 MHz
	h_{FE1} I_C = 25 mA dc		h_{FE2} I_C = 5 mA dc				V_{CE} = 10 V dc I_C = 25 mA dc	V_{CB} = 10 V dc I_E = 0
	min	max	min	max	V dc	V dc		pF
2N5010, 2N5010S U4	30	180	10		1.4	1	1	30
2N5011, 2N5011S, U4	30	180	10		1.5	1	1	30
2N5012, 2N5012S, U4	30	180	10		1.6	1	1	30

Types	h_{FE} at V_{CE} = 10 V dc				$V_{CE(sat)1}$ (1) I_C = 20 mA dc I_B = 5 mA dc	$V_{BE(sat)1}$ (1) I_C = 20 mA dc I_B = 5 mA dc	$\lvert h_{fe} \rvert$ f = 10 MHz	C_{obo} 100 kHz ≤ f ≤ 2 MHz
	h_{FE1} I_C = 20 mA dc		h_{FE2} I_C = 5 mA dc				V_{CE} = 10 V dc I_C = 20 mA dc	V_{CB} = 10 V dc I_E = 0
	min	max	min	max	V dc	V dc		pF
2N5013, 2N5013S U4	30	180	10		1.6	1	1	30
2N5014, 2N5014S, U4	30	180	10		1.6	1	1	30
2N5015, 2N5015S, U4	30	180	10		1.8	1	1	30

(1) Pulsed see 4.5.1.

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Symbol	Dimensions				Note
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.305	.335	7.75	8.51	6
CH	.240	.260	6.10	6.60	
HD	.335	.370	8.51	9.40	
LC	.200 TP		5.08 TP		7
LD	.016	.019	0.41	0.48	8,9
LL	See note 14				
LU	.016	.019	0.41	0.48	8,9
L_1		.050		1.27	8,9
L_2	.250		6.35		8,9
P	.100		2.54		7
Q		.030		0.76	5
TL	.029	.045	0.74	1.14	3,4
TW	.028	.034	0.71	0.86	3
r		.010		0.25	10
α	45° TP		45° TP		7
1, 2, 10, 12, 13, 14					

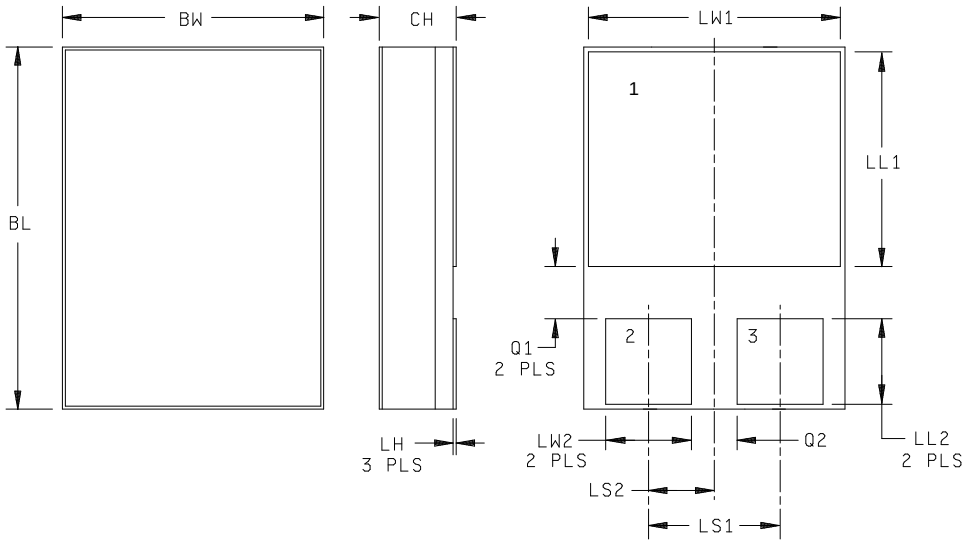


NOTES:

1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Beyond r (radius) maximum, TW shall be held for a minimum length of .011 (0.28 mm).
4. Dimension TL measured from maximum HD.
5. Body contour optional within zone defined by HD, CD, and Q.
6. CD shall not vary more than .010 inch (0.25 mm) in zone P. This zone is controlled for automatic handling.
7. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods or by gauging procedure.
8. Dimension LU applies between L_1 and L_2. Dimension LD applies between L_2 and LL minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
9. All three leads.
10. The collector shall be internally connected to the case.
11. Dimension r (radius) applies to both inside corners of tab.
12. In accordance with ASME Y14.5M, diameters are equivalent to Φx symbology.
13. Lead 1 = emitter, lead 2 = base, lead 3 = collector.
14. For non-S-suffix devices (TO-5), dimension LL = 1.5 inches (38.10 mm) min. and 1.75 inches (44.45 mm) max. For S-suffix types (TO-39), dimension LL = .5 inch (12.70 mm) min. and .750 inch (19.05 mm) max.

FIGURE 1. Physical dimensions (TO-5 and TO-39).

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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.215	.225	5.46	5.72
BW	.145	.155	3.68	3.94
CH	.049	.075	1.24	1.91
LH		.02		0.51
LW1	.135	.145	3.43	3.68
LW2	.047	.057	1.19	1.45
LL1	.085	.125	2.16	3.17
LL2	.045	.075	1.14	1.9
LS1	.070	.095	1.78	2.41
LS2	.035	.048	0.89	1.21
Q1	.03	.070	0.76	1.78
Q2	.02	.035	0.51	0.89
TERM 1	Collector			
TERM 2	Base			
TERM 3	Emitter			

NOTES:
1. Dimensions are in inches.
2. Millimeter equivalents are given for general information only.
3. In accordance with ASME Y14.5M, diameters are equivalent to ∅x symbology.

* FIGURE 2. Physical dimensions, surface mount (U4 version).

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch or https://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

* 2.3 Order of precedence Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and on figures 1 and 2 herein.

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3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in paragraph 1.3, 1.4, and table I.

3.6 Electrical test requirements. The electrical test requirements shall be as specified in table I herein.

3.7 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I and II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

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4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table E-IV MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see E-table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
2	Optional	Optional
3a 3b 3c	Required Not applicable Thermal impedance (transient), method 3131 of MIL-STD-750 (1)	Required Not applicable Thermal impedance (transient), method 3131 of MIL-STD-750 (1)
4	Required	Optional
5	Required	Not applicable
8	Required	Not required
9	I_{CBO1}, h_{FE1}	Not applicable
10	48 hours minimum	48 hours minimum
11	I_{CBO1}; h_{FE1}; ΔI_{CBO1} = 100 percent of initial value or 5 nA dc, whichever is greater. Δh_{FE1} = ±15 percent	I_{CBO1}; h_{FE1}
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein; ΔI_{CBO1} = 100 percent of initial value or 5 nA dc, whichever is greater; Δh_{FE1} = ±15 percent	Subgroup 2 of table I herein; ΔI_{CBO1} = 100 percent of initial value or 5 nA dc, whichever is greater; Δh_{FE1} = ±15 percent
15	Required	Not required
16	Required	Not required

(1) Shall be performed any time after temperature cycling, screen 3a; and does not need to be repeated in screening requirements.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: V_{CB} = 10 - 30 V dc. Power shall be applied to achieve T_J = +135°C minimum using a minimum P_D = 75 percent of P_T maximum, rated as defined in 1.3. With approval of the qualifying activity and preparing activity, alternate burn-in criteria (hours, bias conditions, T_J, mounting conditions) may be used for JANTX and JANTXV quality levels. A justification demonstrating equivalence is required. In addition, the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval. Use method 3100 of MIL-STD-750 to measure T_J.

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4.3.2 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{SW} (V_C and V_H where appropriate). Measurement delay time (t_{MD}) = 70 μs max. See table II, group E, subgroup 4 herein.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein. If alternate screening is being performed in accordance with MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of group A1 and A2 inspection only (table E-VIb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied in accordance with 4.4.2).

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E-VIa (JANS) of MIL-PRF-19500 and 4.4.2.1. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2. See 4.4.2.2 for JAN, JANTX, and JANTXV group B testing. Electrical measurements (end-points) for JAN, JANTX, and JANTXV shall be after each step in 4.4.2.2 and shall be in accordance with table I, subgroup 2.

* 4.4.2.1 Group B inspection (JANS), table E-VIa of MIL-PRF-19500.

Subgroup	Method	Condition
B4	1037	$V_{CB} \geq 10$ V dc.
B5	1027	V_{CB} = 10 V dc; $P_D \geq$ 100 percent of maximum rated P_T (see 1.3). (NOTE: If a failure occurs, resubmission shall be at the test conditions of the original sample.) Option 1: 96 hours minimum sample size in accordance with table E-Via of MIL-PRF-19500, adjust T_A or P_D to achieve T_J = +275°C minimum. Option 2: 216 hours minimum, sample size = 45, c = 0; adjust T_A or P_D to achieve a T_J = +225°C minimum.

4.4.2.2 Group B inspection, (JAN, JANTX, and JANTXV). Separate samples may be used for each step. In the event of a lot failure, the resubmission requirements of MIL-PRF-19500 shall apply. In addition, all catastrophic failures during CI shall be analyzed to the extent possible to identify root cause and corrective action. Whenever a failure is identified as wafer lot or wafer processing related, the entire wafer lot and related devices assembled from the wafer lot shall be rejected unless an appropriate determined corrective action to eliminate the failure mode has been implemented and the devices from the wafer lot are screened to eliminate the failure mode.

Step	Method	Condition
1	1026	Steady-state life: 1,000 hours minimum, V_{CB} = 10 V dc, power shall be applied to achieve T_J = +150°C minimum using a minimum of P_D = 75 percent of maximum rated P_T as defined in 1.3. n = 45 devices, c = 0.
2	1048	Blocking life: T_A = +150°C, V_{CB} = 80 percent rated voltage, 48 hours minimum. n = 45 devices, c = 0.
3	1032	High-temperature life (non-operating), t = 340 hours, T_A = +200°C. n = 22, c = 0.

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4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. For JAN, JANTX, and JANTXV samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot. For JANS, samples shall be selected from each inspection lot. See MIL-PRF-19500.

b. Shall be chosen from an inspection lot that has been submitted to and passed table I, subgroup 2, conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (subgroups B4 and B5 for JANS, and group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

4.4.3 Group C inspection, Group C inspection shall be conducted in accordance with the test and conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and in 4.4.3.1 (JANS) and 4.4.3.2 (JAN, JANTX, and JANTXV) herein for group C testing. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2.

* 4.4.3.1 Group C inspection (JANS), table E-VII of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E; not applicable for U4 devices.
* C5	3131	$R_{\theta JA}$ and $R_{\theta JC}$ only, as applicable (see 1.3).
C6	1026	Test condition B, 1,000 hours at V_{CB} = 10 V dc; power shall be applied to achieve T_J = +150°C minimum and a minimum of P_D = 75 percent of maximum rated P_T as defined in 1.3, n = 45, c = 0.

4.4.3.2 Group C inspection (JAN, JANTX, and JANTXV), table E-VII of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E; not applicable for U4 devices.
C5	3131	$R_{\theta JA}$ and $R_{\theta JC}$ only, as applicable (see 1.3).
C6		Not applicable.

4.4.3.3 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes table I tests herein for conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for C6 life test may be pulled prior to the application of final lead finish. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table II herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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* TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1 2/						
Visual and mechanical examination 3/	2071	n = 45 devices, c = 0				
Solderability 3/ 4/	2026	n = 15 leads, c = 0				
Resistance to solvents 3/ 4/ 5/	1022	n = 15 devices, c = 0				
Temp cycling 3/ 4/	1051	Test condition C, 25 cycles. n = 22 devices, c = 0				
Hermetic seal 4/ 6/ Fine leak Gross leak	1071	n = 22 devices, c = 0				
Electrical measurements 4/		Table I, subgroup 2				
Bond strength 3/ 4/	2037	Precondition T_A = +250°C at t = 24 hrs or T_A = +300°C at t = 2 hrs n = 11 wires, c = 0				
Decap internal visual (design verification) 4/	2075	n = 4 devices, c = 0				
Subgroup 2						
Thermal impedance	3131	See 4.3.2	$Z_{\theta JX}$			°C/W
Collector to base cutoff current	3036		I_{CBO1}			nA dc
2N5010, 2N5010S, U4		V_{CB} = 400 V dc			10	
2N5011, 2N5011S, U4		V_{CB} = 500 V dc			10	
2N5012, 2N5012S, U4		V_{CB} = 580 V dc			10	
2N5013, 2N5013S, U4		V_{CB} = 650 V dc			10	
2N5014, 2N5014S, U4		V_{CB} = 700 V dc			10	
2N5015, 2N5015S, U4		V_{CB} = 760 V dc			10	
Emitter to base cutoff current	3061	V_{EB} = 4 V dc	I_{EBO1}		20	µA dc
Breakdown voltage, collector to base		I_E = 0 mA dc	$V_{(BR)CBO}$			V dc
2N5010, 2N5010S, U4		I_C = 0.1 mA dc		500		
2N5011, 2N5011S, U4		I_C = 0.1 mA dc		600		
2N5012, 2N5012S, U4		I_C = 0.1 mA dc		700		
2N5013, 2N5013S, U4		I_C = 0.2 mA dc		800		
2N5014, 2N5014S, U4		I_C = 0.2 mA dc		900		
2N5015, 2N5015S, U4		I_C = 0.2 mA dc		1,000		
Breakdown voltage, emitter to base		I_C = 0 mA dc, I_E = 0.05 mA dc	$V_{(BR)EBO}$	5		V dc

See footnotes at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Breakdown voltage, collector to emitter	3011	Bias condition D; R_{BE} = 1KΩ I_C = 0.2 mA dc pulsed (see 4.5.1)	$V_{(BR)CER}$			V dc
2N5010, 2N5010S, U4					500	
2N5011, 2N5011S, U4					600	
2N5012, 2N5012S, U4					700	
2N5013, 2N5013S, U4					800	
2N5014, 2N5014S, U4					900	
2N5015, 2N5015S, U4					1,000	
Forward-current transfer ratio	3076		h_{FE1}			
2N5010, 2N5011, 2N5012 2N5010S, 2N5010U4, 2N5011S, 2N5011U4, 2N5012S, 2N5012U4		V_{CE} = 10 V dc; I_C = 25 mA dc		30	180	
2N5010, 2N5011, 2N5012 2N5010S, 2N5010U4, 2N5011S, 2N5011U4, 2N5012S, 2N5012U4		V_{CE} = 10 V dc; I_C = 20 mA dc		30	180	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 5 mA dc	h_{FE2}	10		
Base-emitter saturation voltage	3066	Test condition A; I_B = 5 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)1}$			
2N5010, 2N5010S, U4		I_C = 25 mA dc			1	V dc
2N5011, 2N5011S, U4		I_C = 25 mA dc			1	V dc
2N5012, 2N5012S, U4		I_C = 25 mA dc			1	V dc
2N5013, 2N5013S, U4		I_C = 20 mA dc			1	V dc
2N5014, 2N5014S, U4		I_C = 20 mA dc			1	V dc
2N5015, 2N5015S, U4		I_C = 20 mA dc			1	V dc
Collector-emitter saturation voltage	3071	I_B = 5 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)1}$			
2N5010, 2N5010S, U4		I_C = 25 mA dc			1.4	V dc
2N5011, 2N5011S, U4		I_C = 25 mA dc			1.5	V dc
2N5012, 2N5012S, U4		I_C = 25 mA dc			1.6	V dc
2N5013, 2N5013S, U4		I_C = 20 mA dc			1.6	V dc
2N5014, 2N5014S, U4		I_C = 20 mA dc			1.6	V dc
2N5015, 2N5015S, U4		I_C = 20 mA dc			1.8	V dc

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* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 3 - Continued						
High temperature operation		T_A = +150°C				
Collector to base cutoff current	3036	Bias condition D;	I_{CBO2}			µA dc
2N5010, 2N5010S, U4		V_{CB} = 400 V dc			10	
2N5011, 2N5011S, U4		V_{CB} = 500 V dc			10	
2N5012, 2N5012S, U4		V_{CB} = 588 V dc			10	
2N5013, 2N5013S, U4		V_{CB} = 650 V dc			10	
2N5014, 2N5014S, U4		V_{CB} = 700 V dc			10	
2N5015, 2N5015S, U4		V_{CB} = 760 V dc			10	
Low temperature operation		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 20 mA dc	h_{FE3}	10		
Subgroup 4						
Magnitude of small-signal short-circuit forward current transfer ratio 2N5010, 2N5011, 2N5012 2N5010S, 2N5010U4, 2N5011S, 2N5011U4, 2N5012S, 2N5012U4	3306	V_{CE} = 10 V dc; I_C = 25 mA dc; f = 10 MHz	$\lvert h_{fe} \rvert$	1		
Magnitude of small-signal short-circuit forward current transfer ratio 2N5013, 2N5014, 2N5015 2N5013S, 2N5014S, 2N5015S, 2N5013U4, 2N5014U4, 2N5015U4	3306	V_{CE} = 10 V dc; I_C = 20 mA dc; f = 10 MHz	$\lvert h_{fe} \rvert$	1		
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; f = 2 MHz	C_{obo}		30	pF
Subgroups 5 and 6						
Not required						

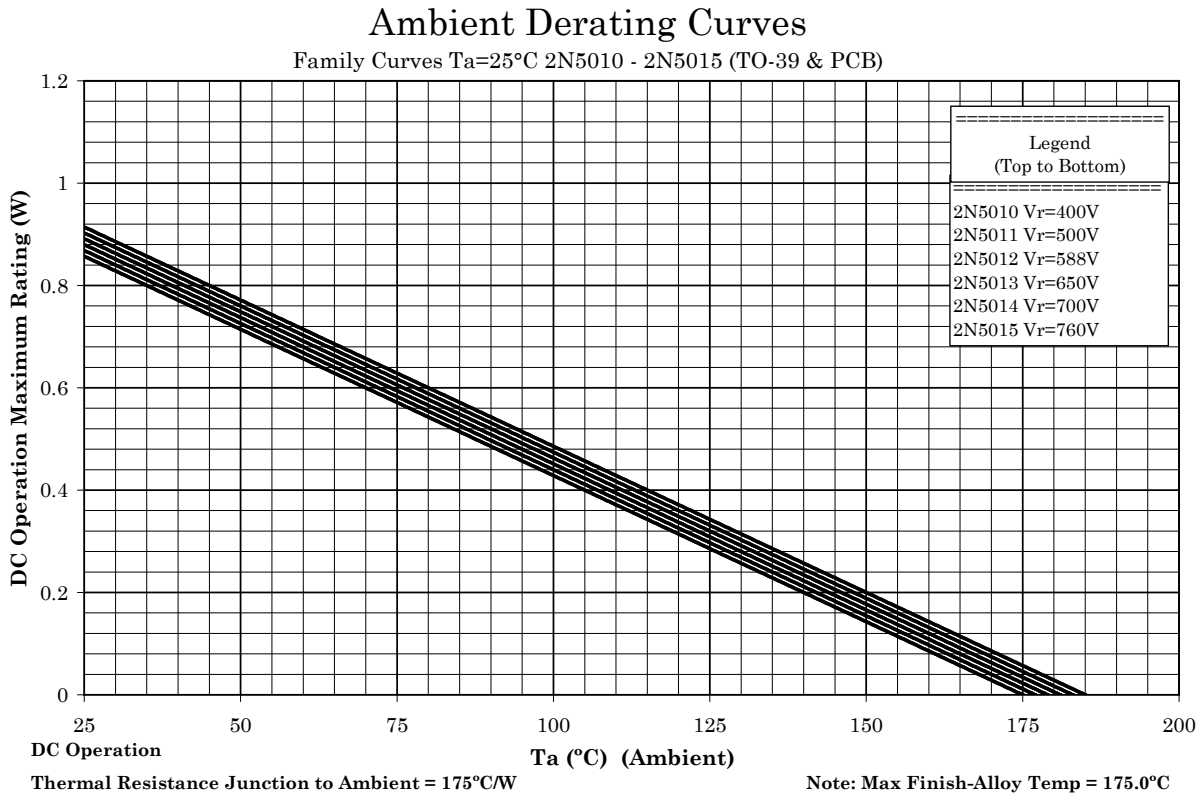
1/ For sampling plan see MIL-PRF-19500.
2/ For resubmission of failed table I, subgroup 1, double the sample size of the failed test or sequence of tests. A failure in table I, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not required for JANS devices.
5/ Not required for laser marked devices.
6/ This hermetic seal test is an end-point to temp cycling in addition to electrical measurements.

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TABLE II. Group E inspection (all quality levels) - for qualification or re-qualification only.

Inspection	MIL-STD-750		Qualification
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling (air to air)	1051	Test condition C, 500 cycles.	
Hermetic seal	1071		
Fine leak Gross leak			
Electrical measurements		See table I, subgroup 2.	
Subgroup 2			45 devices c = 0
Intermittent life	1037	Intermittent operation life: V_{CB} = 10 V dc, 6,000 cycles. Adjust device current, or power, to achieve a minimum ΔT_J of +100°C.	
Electrical measurements		See table I, subgroup 2.	
Subgroup 4			
Thermal impedance		See table E-IX of MIL-PRF-19500, group E, subgroup 4.	
Subgroup 5			15 devices c = 0
Barometric pressure	1001	I_C = 0.1 mA, condition E, pressure = 8 mm HG, normal mounting, t = 60 seconds minimum.	
Subgroup 6			3 devices c = 0
ESD	1020		
Subgroup 8			45 devices c = 0
Reverse stability	1033	Condition A.	

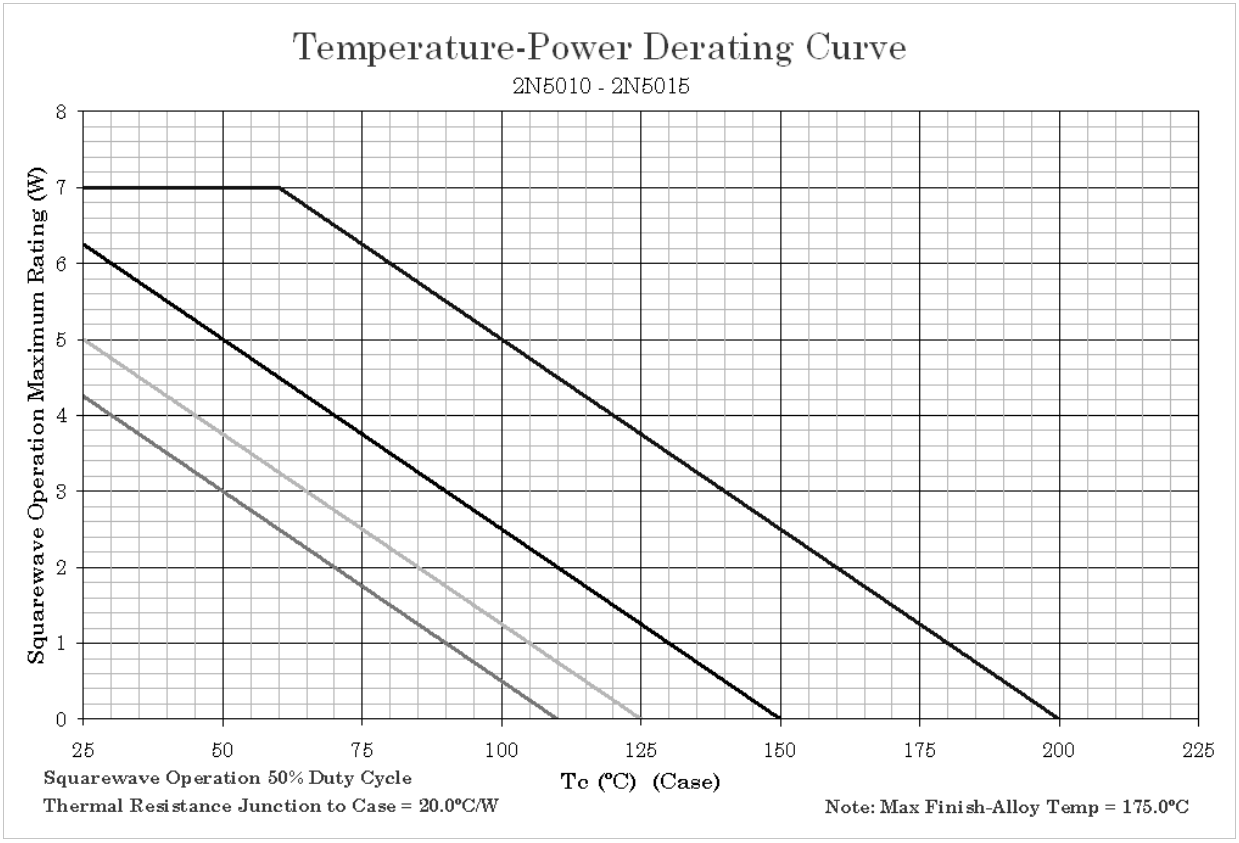
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NOTE: Applies only to ambient (free air) operation either asTO-5, TO-39, or as U4 on a PCB (dc operation).

FIGURE 3. Ambient temperature-power derating curve.

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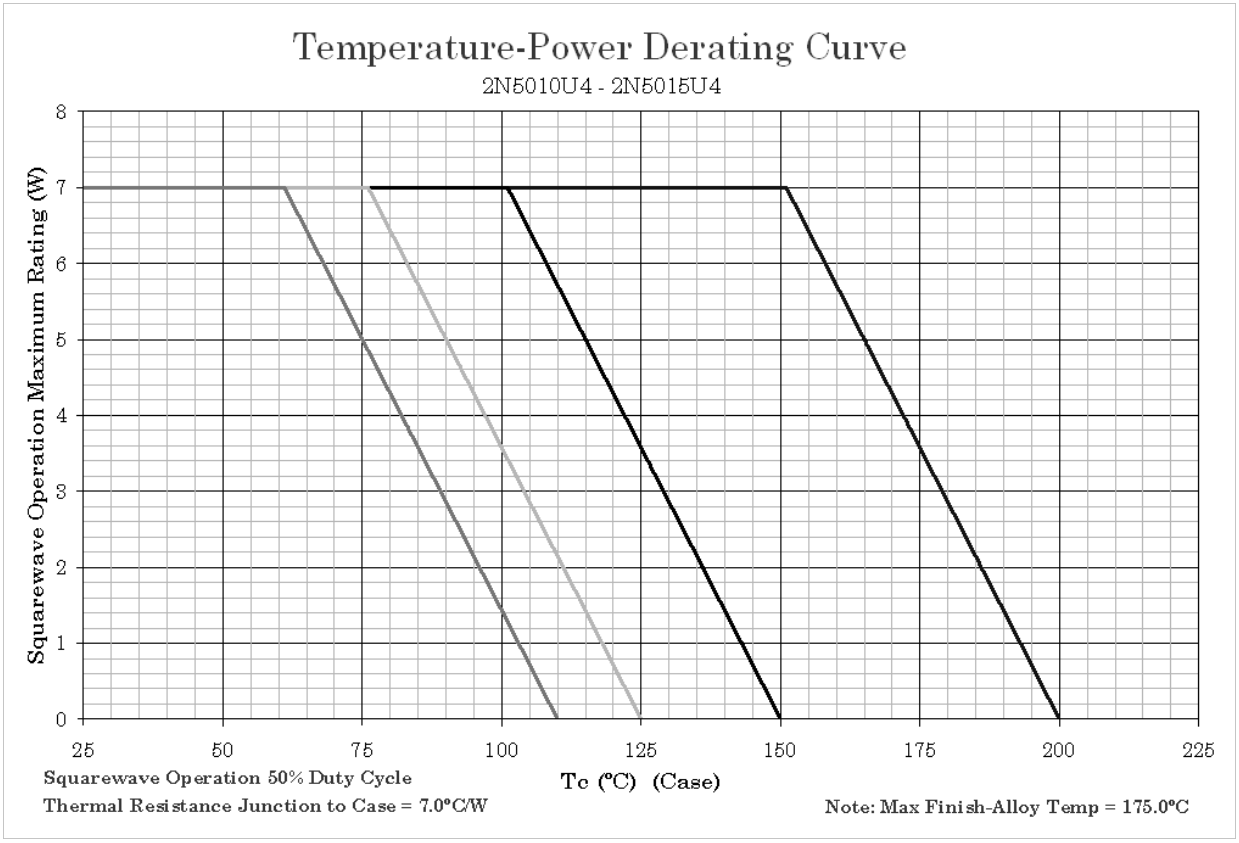


NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature (T_J ≤ 200°C) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at T_J ≤ 150°C, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at T_J ≤, 125°C, and 110°C to show power rating where most users want to limit T_J in their application.

FIGURE 4. Temperature-power derating for steel (TO-5 and TO-39).

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NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature (T_J ≤ 200°C) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at T_J ≤ 150°C, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at T_J ≤, 125°C, and 110°C to show power rating where most users want to limit T_J in their application.

FIGURE 5. Temperature-power derating for (U4).

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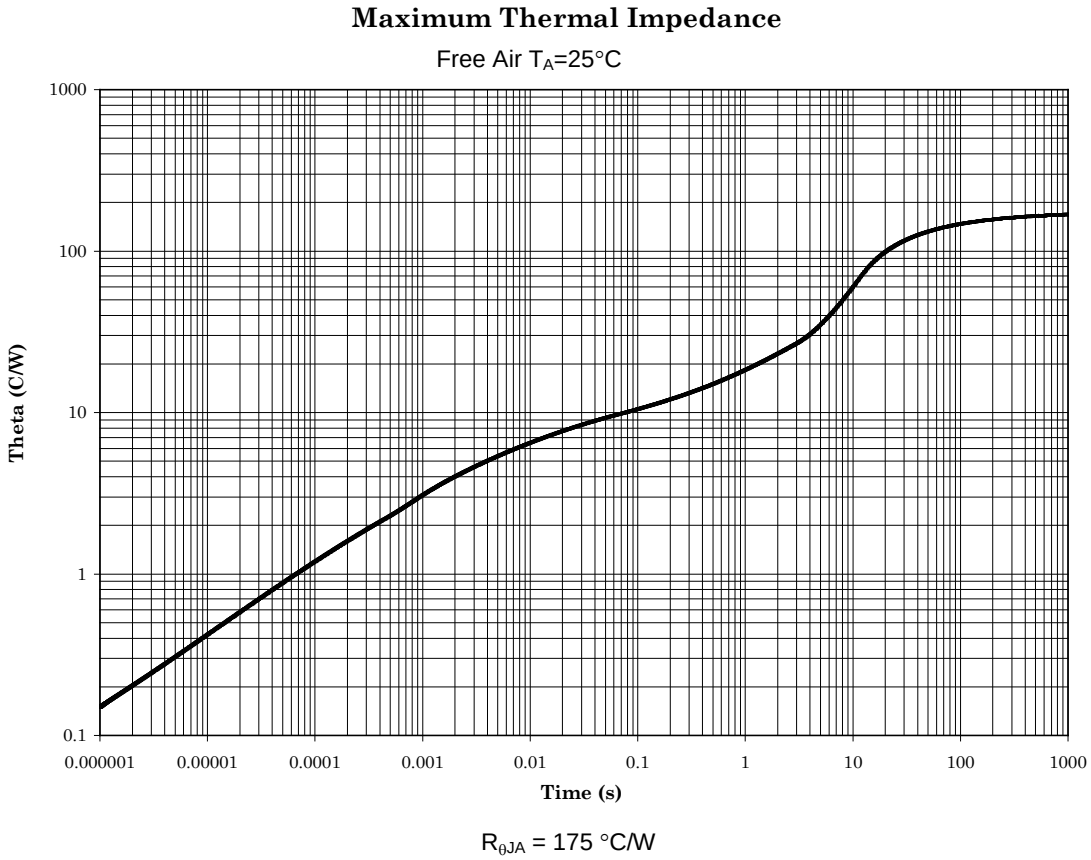


FIGURE 6. Thermal impedance graph for 2N5010 through 2N5015 steel (TO-5 and TO-39).

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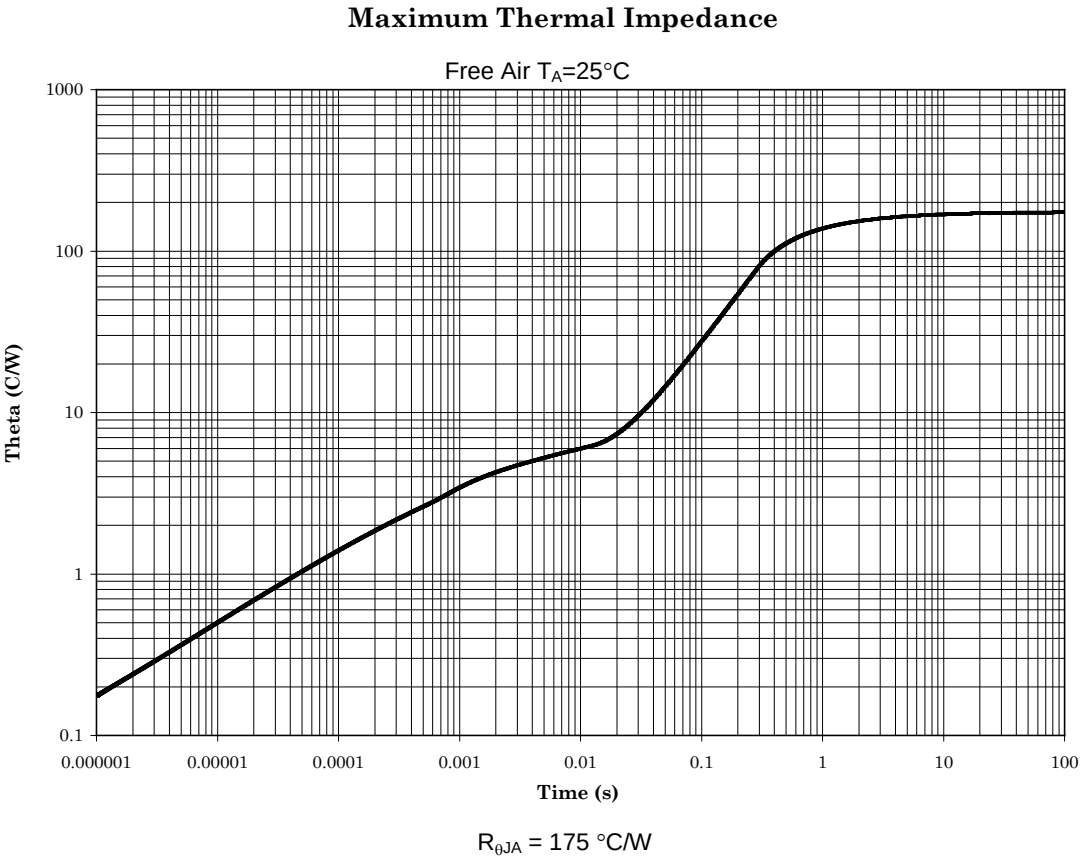


FIGURE 7. Thermal impedance graph for 2N5010U4 through 2N5015U4.

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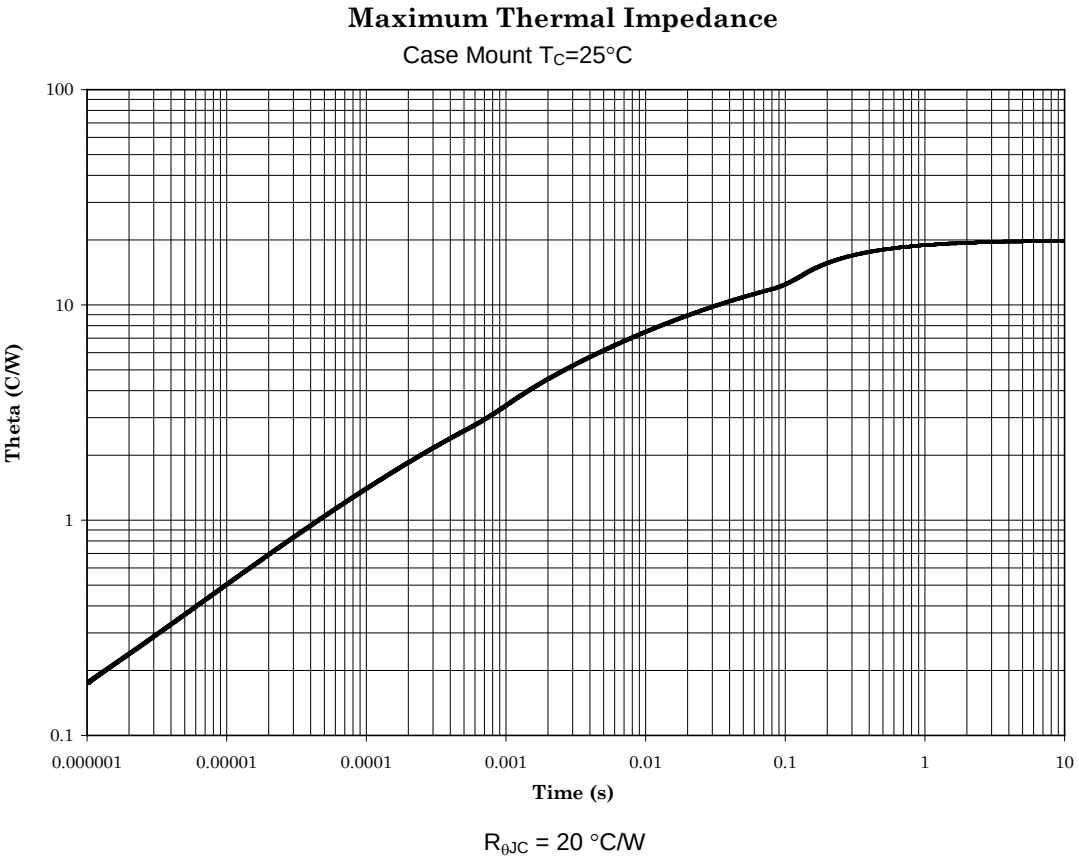


FIGURE 8. Thermal impedance graph for 2N5010 through 2N5015, steel (TO-5 and TO-39).

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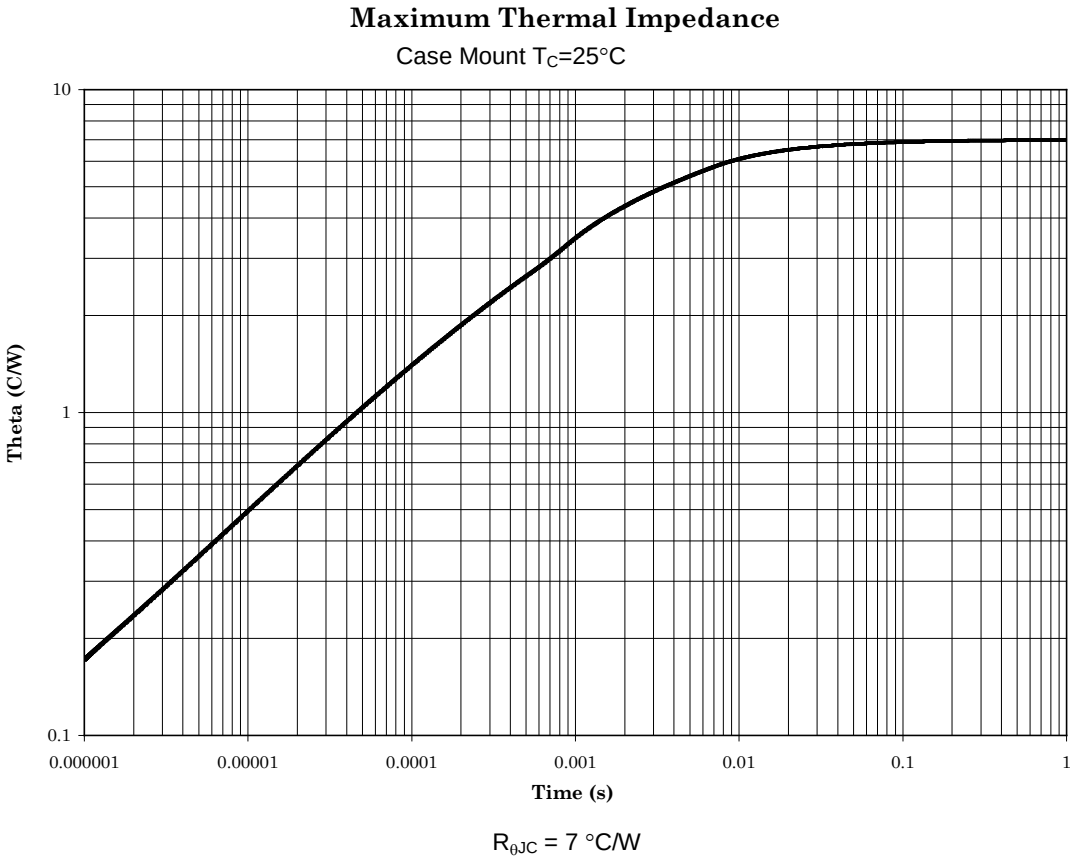


FIGURE 9. Thermal impedance graph for 2N5010U4 through 2N5015U4.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designation

e. Surface mount designation if applicable.

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.daps.dla.mil .

6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, MC
Air Force - 19, 71, 99

Preparing activity:
DLA - CC

(Project 5961-2009-042)

* NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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